GLOBAL ENERGY HACKATHON 2026 · BRIEF 03 · DIGITAL SOLUTIONS & AI

What P90 did for solar finance, Lodestar does for mini-grid demand.

Developers must size a mini-grid before a single meter exists. Today they guess, and lenders can't price a guess.

217,000
mini-grids needed to serve 490 million people by 2030, $127B (ESMAP)

65%
error of survey-based demand estimates per customer (CrossBoundary Innovation Lab)

12.5% → 6.1%
IRR of a 30 kW site when oversized by 100% (CrossBoundary)

None
of the open tools we found combine a demand *range*, a *growth* curve and a lender view, validated against meters

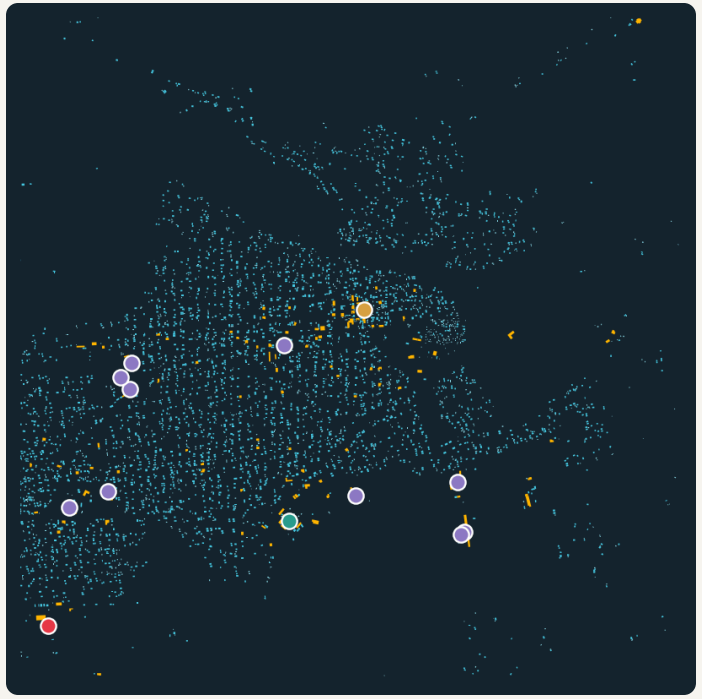
Who: mini-grid developers planning sites in unelectrified villages (Nigeria, Kenya, DRC first). **Who pays:** the lenders, DFIs and results-based-financing programmes that fund them. **Who benefits:** households and businesses that get a system sized to them, at a tariff they can afford, without evening brownouts.

Team Lodestar · Tushar Agarwal

lodestar-utfz.onrender.com · github.com/TusharTechs/lodestar

THE SOLUTION · BUILT THIS WEEKEND

Click a village. Get a bankable demand forecast.

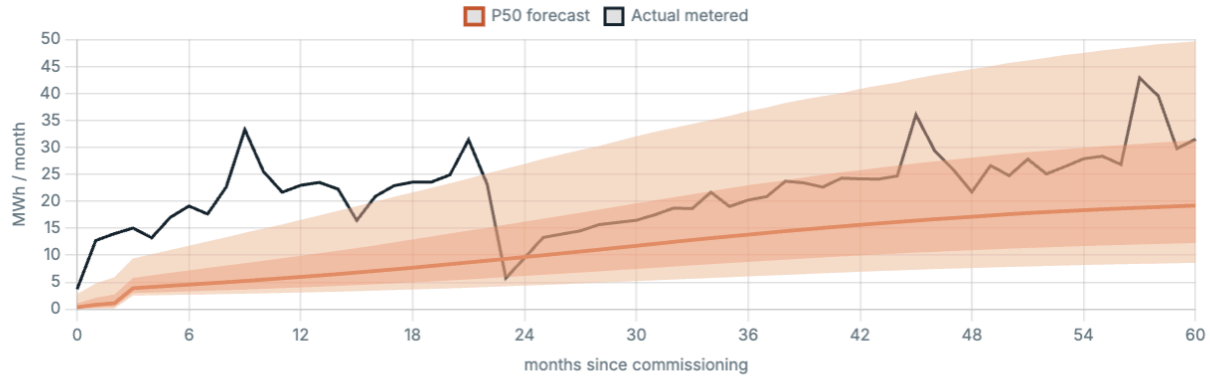


Mutwanga (DRC): 11,400 open building footprints → ~3,800 potential households, 358 businesses; schools, clinic, market, mill from OpenStreetMap.

YEAR-3 DEMAND · P50	P90 · CONSERVATIVE	P10 · OPTIMISTIC	PEAK LOAD · P50
11.9 MWh/mo	**5.3**	**32.4**	**35 kW**
Confidence: **low** · P10/P90 = 6.1×	exceeded with 90% probability	exceeded with 10% probability	load factor 39%

Demand growth, months 0–60 after commissioning

Monthly site energy (MWh). Dark band P75–P25, light band P90–P10. **Black line: actual metered demand**, which this forecast did not see.



Demand Bureau: share the first 6 metered months →

re-weights the forecast with the operator's own meters

Year-3 demand 11.9 MWh/month (P90 5.3, P10 32.4), forecast from footprints and OpenStreetMap alone; it learns only from the other metered sites.

The black line is the site's **real metered demand**, which the model never saw.

Below in the report: driver waterfall, load profile, sizing options, IRR distribution and cash flow at P90/P50/P10.

Range, not a point: P90/P50/P10 from 20,000 Monte Carlo draws

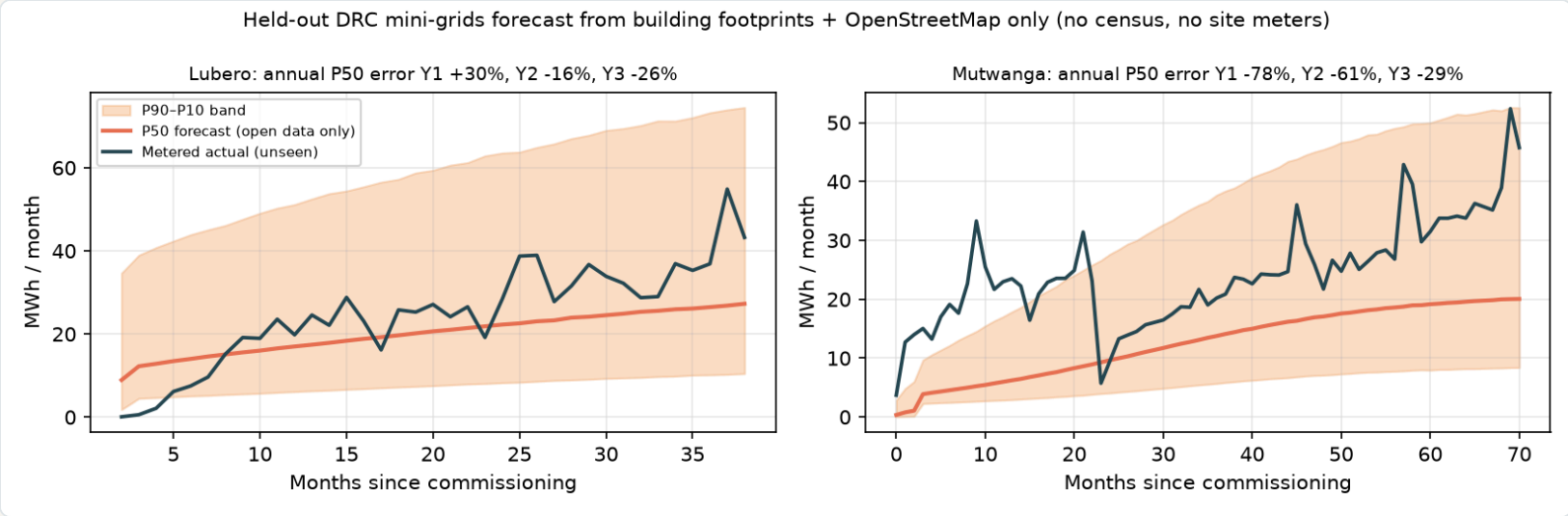
Growth: connections × kWh per connection, months 0–60

Explained in plain words: what each forecast is built from, with every source dated and every driver tagged

Lender view: sizing, grant needed, debt safe on P90, memo, HOMER 8760

PROOF · THE BRIEF'S METRIC: ERROR VS METERED DEMAND ON HELD-OUT SITES

Validated against real meters, using open data alone.



No partner meter data was available, so we found a public CC0 panel: three metered mini-grids in North Kivu, DRC, 2019–2024 (Lunanga et al. 2026). Each village site is forecast with **no census and none of its own meters**; calibration ratios and growth shapes come from the *other* sites only.

5 of 6
site-years of metered energy inside the P90–P10 band (target 80%)

81%
of metered months inside the band

−26% / −29%
year-3 (sizing-year) error, Lubero / Mutwanga · ref. 37% per-site ML (Allee et al.)

Honest limits, all shown in the app: two village sites is method validation; a census-based flat estimate scored 36% on them but gives no range; after 12 metered months, shocks (Mutwanga's conflict collapse) make persistence the better predictor.

Reproduce: python scripts/validate_drc.py

BUSINESS MODEL · WHO PAYS, AND HOW IT PAYS FOR ITSELF

A trust layer the sector can afford, and that improves as it's used.

Key partners

- Mini-grid operators: meter data via the Demand Bureau
- A neutral data steward (e.g. AMDA, which already benchmarks members)
- RBF programmes and rural electrification agencies (e.g. Nigeria REA / DARES)
- Open-data providers: building footprints, OpenStreetMap, NASA POWER, GridFinder

Proposed; none engaged yet.

Key activities

- Forecasts, lender reports, portfolio screens
- Quarterly re-calibration on Bureau data
- Public validation and scoring (ledger)

Key resources

- Open-data pipeline + calibrated model
- Metered validation; forecast ledger
- Bureau data (grows with use)

Value propositions

- **Developers:** a bankable P90/P50/P10 forecast and growth curve *before* building, no survey needed
- **Lenders:** debt sized on P90; stress test across 20,000 scenarios; one-click memo
- **RBF programmes:** screen hundreds of villages: mini-grid, solar home systems, or grid
- **Communities:** right-sized systems: –49% cost of power vs oversizing
- Validated on real meters; open data; explains itself

Customer relationships

- Free forecast → paid report (self-serve)
- **Demand Bureau**: give-to-get, like a credit bureau; site-month aggregates only, published when ≥3 sites from ≥2 operators pool
- Public track record builds trust

Channels

- Web app + API, HOMER export
- USSD / SMS for field officers
- Industry bodies, RBF onboarding, DTECH 2027

Customer segments

- **1 · Primary:** developers bidding into RBF programmes (head most on fire, fast to reach; own the meters). AMDA members plan 6,116 sites
- **2 · Secondary:** RBF programmes (largest budget; Nigeria's DARES targets 1,000+ mini-grids), reached via developers' reports
- **3 · Later:** lenders and DFIs, once the ledger has a track record
- *Beneficiaries, not payers:* households and businesses

Scored on urgency, value, accessibility (team judgement).

Cost structure

Data and licences **$0** · compute **0.76 CPU-s per forecast** (≈$0.00001, measured) · hosting ≈$300/yr · team ≈$60k/yr (two analyst-engineers, Nairobi or Lagos; assumption) · marginal cost per extra site ≈ 0, so cost per forecast falls with scale.

Revenue streams (proposed prices)

Free single-site forecasts · **$300 Bankable Demand Report** per site (0.05% of a $585k mini-grid) · **$12k/yr portfolio screening** for RBF programmes and DFIs. **Break-even ≈ 200 reports/yr**, or 3 subscriptions + 81 reports: 3.3% of AMDA members' pipeline.

Buyer persona	Urgency	Value / ability to pay	Accessibility	Feeds Bureau	Score	Role
Project head at a developer bidding for RBF-funded sites	**5**	**3**	**4**	**5**	**12/15**	**Primary: head most on fire, fast to reach**
Programme manager of an RBF window (e.g. DARES)	4	5	2	5	11/15	Secondary: largest budget, slow procurement
Investment officer at a mini-grid debt fund / DFI	3	4	2	2	9/15	Later: after 12–18 months of scored forecasts

6–18 months: M0–3 backtest 2–3 operators' meters under the Bureau protocol · M3–6 pre-register forecasts for 10 sites · M6–12 score them publicly; pilot with one RBF programme · M12–18 five countries, paid reports to lenders (DTECH 2027 MVP milestone).

Biggest risk: thin metered data. Tested early by the M0–3 operator backtest; mitigated by the Bureau and the ledger.

IMPACT & SCALE · SAME REAL METERS, TURNED INTO DECISIONS

Right-sized, financeable, and reachable from any phone.

−72%
capacity mismatch at year 3 vs a desk-estimate build, replaying real metered demand (155 → 44 kWp, two DRC sites)

+37% → +6%
Mutwanga: desk build oversized; Lodestar's build-at-P50-then-expand plan was within 6%

−49%
cost of power when sized on P50 instead of the optimistic P10 (avg. of 20 villages)

$0.20–0.40
/kWh tariff needed by mini-grid candidates with a Nigeria-NEP-style $600/connection grant, vs ~$1 unsubsidised

Portfolio screener for RBF programmes

20 villages in 15 CPU-seconds: **1** bankable with an RBF grant, **4** need an anchor load, **12** are solar-home-system territory, **3** near the grid. Telling a programme where *not* to build is impact too. Nigeria's DARES alone targets 1,000+ mini-grids.

Built for the field, and for anywhere with rooftops

USSD + SMS: a forecast in one 160-character text on a feature phone. **Offline pack:** 12 KB, opens without internet. Footprints and OpenStreetMap cover all of Africa, South/Southeast Asia and Latin America; a new country needs only its regional benchmark.

Lodestar makes demand bankable, so the next 200,000 mini-grids get built at the right size.

Next: 2–3 operators for the M0–3 backtest, one RBF programme as pilot partner, a neutral host for the Demand Bureau.